LOW-FIDELITY WIREFRAME REPORT · TEXT REPORT

Interview Question Trends: Homepage, Company Page & Compare View

A coding-interview question aggregator whose value is not the list of questions, but the pattern behind them: what a company favours, how that changes over time, and how companies resemble or differ from each other.

> **Design stance.** Start at the overview and let the user drill down — Overview → Company → Topic → Question list. Most aggregators start at the last level. These wireframes invert that: insight first, questions last.

1 — What does a company favour?

Answered by the Company Page: a one-sentence fingerprint, a 100% stacked bar, and a ranked topic table.

2 — How does it change over time?

Answered by sparklines, small multiples, and a topic × month heatmap, all under one global time toggle.

3 — How do companies compare?

Answered by the Homepage matrix and the Compare View: same topics, same order, same scale, up to three companies.

What is in this report

Screen	Question it answers	Page
Homepage	How do companies differ, at a glance?	3–4
Company Page	What is this company’s fingerprint, and how is it shifting?	5–6
Compare View	Where do 2–3 companies align, and where is the gap biggest?	7–8
Shared rules & tests	Metrics, states, data needs, and timed tasks to validate the design	2 and 9

How to read the wireframes

- **Grayscale is deliberate.** Colour would hide hierarchy problems at low fidelity; shading encodes data only — darker means “asks this more than average.” **Numbered markers** in each wireframe link to the annotations on the following page.
- **Example names and numbers** (Meta, Google, Netflix; percentages and counts) are placeholders from the design discussion, showing layout and relative size, not measured data. In the real build, every chart also carries its metric name, time window, and sample size *n*.

Three metrics do all the work

Every screen in this report uses the same three metrics, written the same way. Adding a fourth metric is a design change, not a small feature.

Share	The percentage of a company's questions in the selected window that carry a topic. `share = questions with topic / all questions for that company` Why it matters: raw counts always favour large companies. "42 DP questions" means nothing alone; "19% of Meta's questions" does.
Over-index	How much a company leans into a topic compared with the average company. `over-index = share at company / share across all companies` 1.0× is average. 2.1× means the company asks it at roughly twice the typical rate; 0.5× means it largely avoids it. This is the headline number of the whole product.
Trend	The change in *share* versus the previous equal-length period, in percentage points. `trend = share this period − share previous period` Change in share, not change in count, is what catches a real theme shift — for example a move from Arrays toward Design and concurrency.

Rules that hold on every screen

1. **One primary insight per screen.** One heatmap *or* one ranked list is the hero. Everything else is muted supporting detail.
2. **One colour per topic, everywhere, forever.** If Dynamic Programming is one colour on the homepage, it keeps that colour in trends, on company pages, and in Compare. Cap visible topics at about 12; group the long tail as "Other."
3. **Topic order never silently changes.** The same order appears in the matrix, the ranked table, the stacked bars, and Compare. If a view re-sorts, it says so.
4. **Scales never silently change.** Small multiples and comparison bars share one scale. Re-scaling between companies makes trends lie.
5. **One global time toggle.** Last 30 days / 6 months / 1 year / All time controls everything on the page at once. Default to 6 months — recency is the reason to return.
6. **Show *n* and say when data is thin.** Cells or trends based on fewer than about 10 questions are marked "low data" and are not presented with the same confidence as large samples.
7. **Colour never carries meaning alone.** Every coloured cell or bar also has a number, on the element or on hover / tap.
8. **Do not use** pie charts (cannot be compared), radar charts (look striking, compare badly, and change with axis order), 3-D effects, or a homepage made of nine competing widgets.

> **Accessibility baseline.** Use a colourblind-safe palette when colour is added, keep contrast high against shaded cells, and keep matrix headers sticky so users do not lose context while scrolling.

Homepage — how companies differ

LOGO Companies Topics Trends Compare **[Search] (1)**

What are companies actually asking?

Patterns across 12,483 questions · Updated Oct 19

Time: 30d | **6m** | 1y | All [x] Weight by recency (2)

COMPANY × TOPIC — darker = asks this MORE than average (3)

	Array	Graph	DP	Tree	Design	Heap	Cmp
Meta	0.9×	1.8×	2.1×	1.5×	1.1×	0.8×	[ ]
Google	1.1×	1.4×	2.3×	1.3×	0.7×	1.2×	[ ]
Amazon	1.7×	1.8×	1.2×	2.2×	0.5×	1.6×	[ ]
Netflix	0.8×	1.2×	0.5×	0.6×	3.4×	·	[ ]
Goldman	2.5×	0.3×	1.2×	·	0.7×	1.3×	[ ]

Legend: pale = under-indexed · mid = average · dark = 2×+ average | Hover a cell: “Meta · DP: 19% of questions, 2.1× avg, n=34” | Top ~10 companies × top 12 topics only; sticky headers.

BIGGEST MOVERS (vs previous 6m) (4)

↑ OpenAI — Design +9 pts
↓ Google — Strings −6 pts
↑ Netflix — Graphs +5 pts
[See all trends →]

MOST DISTINCTIVE (5)

1. Netflix — Design-heavy
2. Goldman — Arrays / Math-heavy
3. Meta — Graphs / DP-heavy

Click a row → company page

RECENT QUESTIONS (secondary) (6)

Company ▾ | Topic ▾ | Difficulty ▾

Question title … Meta · Graphs · Med · 3 days ago
Question title … Google · DP · Hard · 5 days ago

Sticky compare tray (7) — appears only after a user checks 2+ companies:

Compare: [Meta ×] [Netflix ×] [Compare now →]

Homepage — annotations

1. **Search is not the main event.** Search finds a company or a question, but a first-time visitor gets value from the matrix without typing anything. Do not gate the insight behind a search box.
2. **One time toggle controls everything.** The headline count, the matrix, movers, and the question list all use the same window. Charts on one page must never silently use different windows.
3. **The matrix is the hero, and the only hero.** Rows are companies, columns are the top 12–15 topics, and colour encodes over-index on a diverging scale (pale = under, saturated = over). People spot clusters in it instantly — for example product companies leaning into Trees and Graphs while banks lean into Arrays and Strings — without reading a single number. Cells based on fewer than 10 questions get a hatch / “low data” treatment instead of a confident colour.
4. **Movers are change in share, in percentage points.** A company can ask more DP questions simply because it asked more questions overall. Change in share is the signal that a theme is actually shifting.
5. **Distinctiveness is the non-chart entry point.** Some visitors will not read a heatmap. A plain ranked list — ranked by how far a fingerprint sits from the average — gives them the same insight in words, and every row is a route to a company page.
6. **The question list exists, but below the fold.** It is filterable by company, topic, and difficulty, and it is deliberately not the hero. Leading with it would make the product a list site like every competitor.
7. **The compare tray is earned, not permanent.** It appears only after 2+ companies are checked, holds a maximum of 3, and carries the selection into the Compare View (pages 7–8) without re-selecting.

Mobile behaviour

Do not shrink the matrix to fit a phone screen — it becomes unreadable long before it becomes small. Instead, the mobile homepage defaults to a company picker plus that company’s ranked topic bars (the company page pattern, page 5). The full matrix remains available as a horizontally swipeable table with the company column pinned.

Empty and thin-data states

- **New or small company:** show the fingerprint with its real *n* and a “low data — interpret with care” label, rather than hiding the company.
- **No movement to report:** if no mover clears a minimum change threshold, say “No large shifts this period” instead of promoting noise into a headline.
- **First visit, no selection:** the page is complete as drawn. Nothing on the homepage requires an account, a search, or a selection first.

> **Success test for this screen:** a new visitor can answer “Which of these companies leans most into Design?” in under 10 seconds, from the matrix alone.

Company Page — one company's fingerprint

← All companies / **Meta**

META n = 184 · last 6m (1)

"Leans heavily into Graphs, DP and Trees. Light on Strings and Math vs. average."

Similar: [Google] [Netflix] Different: [Goldman] [Oracle] (2)

Time: [30d] [6m] [1y] [All] [+ Compare]

FINGERPRINT — share of Meta's questions (3)



Graphs 24% · DP 19% · Trees 16% · Arrays 12% · Heap 7% · Other 22% — 100% stacked bar; hover a segment for % + count.

TOPICS RANKED (4)

Topic	Share	vs Avg	Trend	Change	
1 Graphs	24%	1.8×	/\/~	+4 pts	[Questions →]
2 DP	19%	2.1×	/~~	+2 pts	[Questions →]
3 Trees	16%	1.5×	~_	−1 pt	[Questions →]
4 Arrays	12%	0.9×	_/~/	±0	[Questions →]
5 Heap	7%	0.8×	~~	−2 pts	[Questions →]

[Show all 18 topics ▾] — top 5 shown by default; long tail grouped under "Other" above.

HOW IT'S CHANGED (5)

[Small multiples] [Topic × Month heatmap]

Graphs



share rising — May … Oct

DP



share falling — May … Oct

In the build these are true small-multiple line panels, one per top topic, on one locked y-scale.

QUESTIONS — Graphs (filtered by the section above) (6)

[Difficulty ▾] [x Recent first]

- Graph traversal variant · Med · 12 Oct · seen 4×
- Course schedule variant · Med · 28 Sep · seen 3×

Company Page — annotations

1. **The auto-summary is the most important element on the page.** One plain-English sentence, generated from the top two over-indexed topics and the top under-indexed topic, next to the sample size and window. A visitor who reads nothing else still leaves with the answer.
2. **Similarity sits at the top, not in a footer.** “Most similar” and “least similar” chips, computed from the similarity of topic-share vectors, are navigation shortcuts to the companies a visitor would compare next anyway. Each chip can also seed the Compare View directly.
3. **Bar plus table: the whole, then the precision.** The 100% stacked bar shows the shape of the fingerprint at a glance but cannot be read precisely; the ranked table below gives exact share, over-index, and change. Neither works alone — keep both, in that order.
4. **The ranked table is the workhorse.** Sparklines carry shape only and deliberately have no axes; the Change column carries the number. Default to the top 5 topics, expand to all topics on request, and keep the topic order here identical to the homepage matrix and to Compare.
5. **Trends: small multiples by default, heatmap on toggle.** One mini-panel per top topic, all on the same y-scale, is easier to read than one multi-line chart with 20 series. The topic × month heatmap (rows = topics, columns = months, colour = share) is the power-user alternative, offered as a view toggle on the same data — not a separate page.
6. **Selecting a topic filters in place.** Clicking a topic row does not navigate away; it filters the question list at the bottom of the same page, preserving the fingerprint above it as context.

Worked reading of the example

For the placeholder Meta example: Graphs is both the largest share (24%) and rising fastest (+4 points), while DP is the most distinctive topic (2.1× the average) even though it is second by share. That distinction — biggest versus most distinctive — is exactly why Share and Over-index are shown as separate columns rather than merged into one score.

Thin-data behaviour

- **30-day window with very few questions:** if *n* falls below the low-data threshold, show an inline prompt — “Only 6 questions for this company in 30 days” — with a one-click button to widen to 6 months. Never show a confident trend built on a handful of questions.
- **Missing topic:** a topic with no questions in the window shows 0% and 0.0×, not a blank — an absence is itself information about what a company does not ask.

> **Success test for this screen:** a visitor can answer “What does Meta ask more than most companies, and is that increasing?” in under 10 seconds, without scrolling.

Compare View — where companies align and diverge

← Home / **Compare companies**

Meta × | Google × | Netflix × | + Add company (max 3) (1)

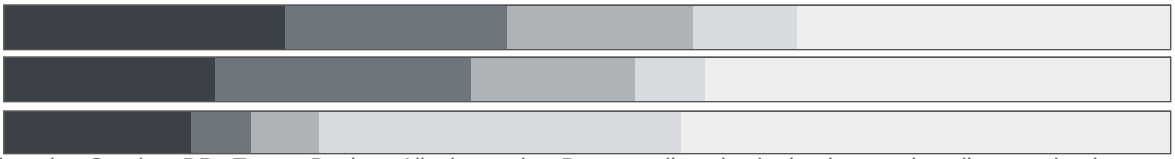
Time: 30d | **6m** | 1y | All n: Meta 184 · Google 231 · Netflix 96

VERDICT (2)

“Meta and Google are close — both lean into Graphs and DP. Netflix is the outlier: Design-heavy, DP-light.”

Meta–Google	Meta–Netflix	Google–Netflix
82% similar	**41% similar**	**38% similar**

FINGERPRINTS — same segment order, left to right (3)



Segments in order: Graphs · DP · Trees · Design · All other topics. Bars are aligned at both edges — that alignment is what makes them comparable.

TOPICS, ALIGNED — one row per topic, one shared scale (4)

Topic	Meta	Google	Netflix	Gap
Design	9% · 1.1×	6% · 0.7×	**31% · 3.4×**	**25 pts**
DP	19% · 2.1×	22% · 2.3×	5% · 0.5×	17 pts
Trees	16% · 1.5×	14% · 1.3×	6% · 0.6×	10 pts
Graphs	24% · 1.8×	18% · 1.4×	16% · 1.2×	8 pts
Arrays	12% · 0.9×	15% · 1.1×	10% · 0.8×	5 pts

Sort: **Biggest gap first** | Global topic order — each cell also carries a bar on the same 0–35% scale in the build. [Show all topics ▾]

SHARED LEANINGS (5)

All three: Arrays close to average (0.8–1.1×).
Meta + Google: Graphs and DP above average.

WHERE THEY DIVERGE (5)

Netflix alone: Design 3.4× average.
Meta + Google ask DP; Netflix largely does not.

QUESTIONS — Design (largest-gap topic, selected above) (6)

Show: all three companies | Recent first

- System design variant · Netflix · Hard · 9 Oct
- Design a rate limiter · Meta · Med · 2 Oct

Compare View — annotations

Comparison is where most designs in this category fall apart: each company gets its own chart, its own topic order, and its own scale, and the reader is left doing the comparison mentally. Every choice below exists to prevent that.

1. **Maximum three companies, enforced in the interface.** Two is the clearest comparison; three is still readable. Beyond three, aligned bars and tables stop being scannable — the “add” control disables at three and says why. Selections arrive pre-filled from the homepage compare tray or a company page’s similarity chips, and the selection is encoded in the URL so a comparison can be shared or bookmarked.
2. **Lead with a verdict, not with charts.** A generated sentence names the closest pair and the outlier, followed by the three pairwise similarity scores (similarity of the topic-share vectors, expressed as a percentage). The reader gets the conclusion first, then the evidence below it.
3. **Stacked fingerprints, stacked vertically.** One 100% bar per company, one above the other, with segments in identical left-to-right order and identical colours. Vertical stacking with shared edges is comparable; three bars side by side in separate cards, or a radar chart, are not.
4. **The aligned topic table is the core of the screen.** One row per topic, one column per company, every bar drawn on a single shared scale, and each cell showing both share and over-index. The final column — **Gap**, the difference in share between the highest and lowest company — is the ranking signal for the whole view, and the default sort is “biggest gap first.” The alternative sort, global topic order, keeps the order used on every other screen. Companies are never sorted independently of each other.
5. **Shared versus divergent, in words.** Two short lists sit under the table: what the selected companies have in common, and where exactly one of them stands apart. This is the fastest route to the “similarities” half of the original question, and it requires no chart literacy.
6. **Questions follow the gap.** The question list at the bottom defaults to the largest-gap topic, so the reader sees concrete questions behind the biggest difference first. Selecting any other table row re-filters the list in place.

States that must be designed, not improvised

State	Behaviour
Only 1 company selected	Show that company’s fingerprint with an empty, clearly labelled second slot: “Add a company to compare.” Suggest its most-similar and least-similar companies as one-click additions. Do not render a comparison with one side missing.
Unequal sample sizes	Show each company’s *n* in the header at all times. If one company’s *n* is below the low-data threshold, mark its column “low data” throughout and soften the verdict wording (“based on limited data”).
Removing a company	Removing the middle company keeps the remaining two in place and re-computes verdict, gaps, and similarity instantly — no page reload, no loss of the time window.
Trend in Compare (optional, v2)	If trends are added, chart only the top 3 divergent topics, one small panel per topic with one line per company on a shared scale. Never overlay all topics in one chart.

Data requirements and validation tests

Data requirements behind these wireframes

Requirement	Why the wireframes depend on it
One clean topic taxonomy, with a hierarchy (e.g. Arrays > Two Pointers)	Messy tags produce messy visualisations no layout can fix. Cap visible topics at about 12–15; roll the rest into “Other.”
A date on every question	Trend and movers need a time window plus an equal-length previous period to compare against.
Share and over-index, precomputed per company × topic × window	Every screen reads the same aggregates; per-page computation guarantees disagreement.
Sample size stored with every aggregate	The low-data treatment on all three screens is driven by *n*.
Pairwise similarity between fingerprints	Powers the company page chips, the distinctiveness ranking, and the Compare verdict — one computation, three surfaces.

Validate with timed tasks, not opinions

Test with five people: give tasks, time them, and simplify any view where the answer takes over about 10 seconds.

1. “Which company asks more Dynamic Programming — Meta or Amazon?” (Matrix, then Compare)
2. “Is Netflix asking more or less Graphs than last year?” (Company page trend)
3. “What is Meta’s most distinctive topic — not its biggest, its most distinctive?” (Over-index column)
4. “Which two of these three companies are most alike, and on what?” (Compare verdict)
5. “Where is the single biggest difference between Meta and Netflix?” (Compare, Gap column)

Suggested build order

1. **Lock the taxonomy and metric definitions first,** then build the aggregates (share, over-index, trend, similarity) as one shared data layer, with *n* attached.
2. **Build the Company Page next** — the simplest complete loop, and it validates the metrics first. **Then the Homepage matrix, then Compare,** reusing the same aggregates, order, and scale.
3. **Add colour last,** one colour per topic, identically everywhere, once the grayscale hierarchy is proven.

> **In one sentence:** show share instead of counts, over-index instead of raw popularity, trends as change in share, and comparisons only on a shared order and scale — with the question list last, not first.